

Lab 05

Object Oriented Programming

BS DS/CY Fall 2025 Morning/Afternoon

Instructions:

- Indent your code properly.
- Use meaningful variable names and follow standard naming conventions.
- Use clear and meaningful prompts/labels for all program input and output.
- Make sure that there are NO dangling pointers or memory leaks in your program.
- Do not discuss your problems with fellow students. If you face any difficulty in understanding, consult your TAs.

Task # 1

Write a Circle class that has the following member variables:

- **radius:** a double
- **pi:** a double initialized with the value 3.14159

The class should have the following member functions:

- **Default Constructor.** A default constructor that sets radius to 0.0.
- **Constructor.** Accepts the radius of the circle as an argument.
- **setRadius.** A mutator function for the radius variable.
- **getRadius.** An accessor function for the radius variable.
- **getArea.** Returns the area of the circle, which is calculated as
$$\text{area} = \text{pi} * \text{radius} * \text{radius}$$
- **getDiameter.** Returns the diameter of the circle, which is calculated as
$$\text{diameter} = \text{radius} * 2$$
- **getCircumference.** Returns the circumference of the circle, which is calculated as
$$\text{circumference} = 2 * \text{pi} * \text{radius}$$

Write a program that demonstrates the Circle class by asking the user for the circle's radius, creating a Circle object, and then reporting the circle's area, diameter, and circumference.

Task # 2

Create a class **Date** that has 3 private member variables to store following three attributes:

- **Day** (an integer)
- **Month** (an integer)
- **Year** (an integer)

***Note:** All member functions of the **Date** class, which are not supposed to change data stored in the calling object, should be declared as **const** member functions.*

- Implement the getter and setter functions for each of the above three member variables.
Names of these functions should be: **setDay**, **setMonth**, **setYear**, **getDay**, **getMonth**, **getYear**.
- Implement a public member function **displayDate** of the **Date** class. This function should display the date in the following format. For example, the date 15th March 2016 should be displayed on screen like **15/03/2016**
- Write a main function (driver program) to test the working of **Date** class and its member functions.
- ~~In the main function, allocate an array of 5 **Date** objects. Store these five dates in these **Date** objects: 25th December 1876, 9th November 1877, 21st April 1938, 14th August 1947, and 11th September 1948. Now, print these five dates by calling the appropriate member function of the **Date** class.~~
- ~~Proceed as in part (d) above, but now the array of dates should be allocated dynamically.~~

~~Your program should ask the user about the size of the array and then allocate the array dynamically. The program should then ask the user to enter those many dates and store them into the array. Finally, your program should display all dates on screen.~~

We will discuss part d and e in coming week.

Task # 3

Create a class **SaleItem** that a grocery store might use to represent a particular item being sold at the store. The class **SaleItem** should have private member variables to store each of the following:

- **ID** (an integer to uniquely identify each item)
- **Name** (a char array of size 41)
- **Quantity** (an integer to represent how many such items are present in the store)
- **Price** (a double value indicating the price of the item in Rupees)

*Note: All member functions of the **SaleItem** class, which are not supposed to change data stored in the calling object, should be declared as **const** member functions.*

- Implement the **getter and setter functions** for each data member of the **SaleItem** class.
- Implement a public member function **incrementQuantity** which will take an integer value as argument and increments the quantity of the calling **SaleItem** object by that much.
- Implement a public member function **decrementQuantity** which will take an integer value as argument and decrements the quantity of the calling **SaleItem** object by that much.
- Implement a public member function **displayInformation** of the **SaleItem** class. This function should display all the information about the calling **SaleItem** object on the screen, in a neat and readable way.
- Write a main function (driver program) to test the working of **SaleItem** class and its member functions.
- ~~Write a menu based program using the **SaleItem** class. Your program should ask the user that how many **SaleItems** he/she wants to create. Then, it should dynamically allocate an array of **SaleItems**. Then, the program should ask the user to enter the values of all attributes for each **SaleItem** object (see the input format in the following sample run).~~

~~After that a menu should be displayed to the user, showing the ID, Name, Quantity, and Price of each item. The user should be able to select (buy) any **one** item by specifying its **ID** and the **quantity** that he/she needs to buy. After that, the program should display the bill of the user (see the output format in the following sample run).~~

~~Make sure that your program **decrements** the quantity of the **SaleItem** which was bought by the user. After displaying the bill the user should be asked to press any key to return back to the menu. The program should eventually terminate when the user enters **-1** at the menu. At the end, make sure that all the dynamically allocated memory is properly deallocated.~~

*Note: In the following sample run, text that is shown in **red** is entered by the user.*

We will discuss **part f** in coming week.

How many SaleItems you want to create: 3

Enter details of SaleItem # 1: 901,Tuc Biscuits,150,15

Enter details of SaleItem # 2: 808,Lays Chips,100,20

Enter details of SaleItem # 3: 850,Dairy Milk Choc,200,10

Menu

901	Tuc Biscuits	150	15
808	Lays Chips	100	20
850	Dairy Milk Choc	200	10

Enter the ID of item that you want to buy (-1 to exit): 901

Enter the Quantity you want to buy: 6

Your Bill is as follows:

ID	Name	Quantity	Unit Price
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901	Tuc Biscuits	6	15
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Total Amount: Rs.90

Press any key to return to menu: ↵

Menu

901	Tuc Biscuits	144	15
808	Lays Chips	100	20
850	Dairy Milk Choc	200	10

Enter the ID of item that you want to buy (-1 to exit): -1

Bye Bye!!